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Project Report

Project: Six-State Medicaid Provider Exclusion List ETL **Project Period:** 06/15 - 06/28
GitHub Repository: <https://github.com/Xinzhuo-Li/medicaid-exclusion-list>

Subject of This Project

Six-State Medicaid Exclusion List Merge — ETL Pipeline, PostgreSQL Deployment, Data Quality Validation, and GitHub Documentation

- Built an end-to-end ETL pipeline to merge six assigned state Medicaid exclusion lists (MD, MA, MI, MS, MT, NE) into a unified OIG LEIE-compatible schema.
- Converted source Excel and PDF files into standardized cleaned CSV outputs and loaded them into PostgreSQL on the vesta server.
- Deployed 8,575 cleaned exclusion records to PostgreSQL tables: six stage tables, cleaned_staging, and exclusion_main.
- Resolved state-specific data mapping issues including Michigan dual sanction dates, Mississippi indefinite REINDATE values, Maryland legend rows, and Montana alias names.
- Documented leadership decisions on full-name retention, empty EXCLDATE handling, cross-state NPI retention, and strict main/staging sync.
- Added automated regression tests (55 pytest cases), pipeline quality audit, run manifest tracking, and GitHub onboarding documentation.
- Published the project to GitHub with README, WORKFLOW runbook, and reproducible source files for team use.

Notes

1. Project Purpose and Scope

The project supports the EMRTS Medicare/Medicaid Provider Exclusion Database initiative by consolidating state Medicaid exclusion lists into one searchable PostgreSQL database.

Assigned scope covers six states: Maryland (MD), Massachusetts (MA), Michigan (MI), Mississippi (MS), Montana (MT), and Nebraska (NE).

Each state's raw data format differs (Excel vs PDF, different column names and business rules), so per-state converters were required before merging into a common schema.

The target schema follows the OIG LEIE reference (18 fields plus source_state), enabling consistent provider screening by name, NPI, business name, state, and exclusion date.

Final deliverable: 8,575 cleaned records loaded and merged into exclusion_main, with full audit trail and documented business decisions.

2. Technologies and Tools Used

Python 3.12 — core ETL language with modular per-state converters and shared cleaning utilities.

pandas / openpyxl — Excel source file parsing for five states.

pdfplumber — Nebraska PDF table extraction (no OCR required).

PostgreSQL — production database on vesta (port 5433); stage tables, cleaned_staging, and exclusion_main.

psycopg2 — Python PostgreSQL loader; SQL scripts for schema, merge (DELETE + INSERT), and sync verification.

pytest — 55 automated regression tests for dates, NPI, names, dedup, Michigan multi-date, and MS REINDATE logic.

Git / GitHub — version control and public repository: <https://github.com/Xinzhuo-Li/medicaid-exclusion-list>

Linux server (vesta) — remote deployment via rsync/SSH; PostgreSQL setup and merge scripts.

Documentation — README.md, docs/WORKFLOW.md, docs/DATA_INVENTORY.md,

docs/STATE_MAPPING.md, docs/ISSUES_AND_DECISIONS_BILINGUAL.md.

3. Pipeline Architecture (Steps 0–5)

Step 0 — Run manifest: records SHA256 hashes of raw source files for auditability (docs/run_manifest_*.json).

Step 1 — Convert & clean: per-state modules read data/raw/ `xlsx/pdf` and write data/processed/*_raw.csv and data/cleaned/*_oig.csv.

Step 2 — Import validation: check row counts, NPI/date rates, and field lengths against baselines; fail-fast on drift.

Step 3 — Quality audit: rerun converters and compare to saved CSVs for rerun consistency; fail-fast on mismatch.

Step 4 — Load PostgreSQL: apply schema, truncate and reload stage tables and cleaned_staging.

Step 5 — Merge & verify: DELETE six-state rows from exclusion_main, INSERT from cleaned_staging, assert strict EXCEPT sync.

Orchestrator: `python3 -m src.pipeline` (flags: `--skip-nebraska`, `--skip-db`, `--skip-merge`).

4. Data Processing Results by State

MD: 1,605 source rows → 1,603 cleaned (6 legend rows filtered, 2 exact duplicates removed).

MA: 294 → 294 (no address in source Excel).

MI: 3,982 → 4,921 (multi-date expansion — each sanction date becomes a separate record).

MS: 194 → 193 (1 empty row skipped; indefinite REINDATE cleared for 192 records).

MT: 174 → 174 (a.k.a. alias names parsed; 6 records missing EXCLDATE in source).

NE: 1,391 → 1,390 (PDF extraction; 82% missing NPI; no address column in source).

Total: 7,640 source rows → 8,575 cleaned records in exclusion_main.

5. Issues Resolved During Development

Michigan dual sanction dates: previously mapped incorrectly; fixed so each parsed date generates a separate record with its own EXCLDATE and EXCLTYPE.

Mississippi indefinite REINDATE: Excel serial 401768 and 'Indefinite' exclusion period now map to empty REINDATE instead of a placeholder date.

Maryland legend rows: filtered metadata rows that were not provider records.

Montana alias names: parsed 'a.k.a.' prefix names into last/first/middle fields.

Deduplication: exact-duplicate removal using 16 identity fields; avoids collapsing different individuals with similar names.

EXCLTYPE normalization: state-specific sanction codes mapped to standardized values where applicable.

6. Leadership Decisions (Confirmed 2026-06-20)

Decision 1 — Retain full name length: name fields use PostgreSQL TEXT; no OIG fixed-length truncation (272 records with busname > 30 chars preserved).

Decision 2 — Empty EXCLDATE = long-term exclusion: 8 records (MT 6 + MS 2) kept with empty excldate; no placeholder dates inserted.

Decision 3 — Cross-state duplicate NPI: 15 NPIs appearing in multiple states retained as separate rows with source_state for audit.

Decision 4 — Strict sync: exclusion_main must exactly match cleaned_staging after merge; verified by sql/04_verify_main_sync.sql and Python assert.

7. Known Source Data Limitations

Nebraska: 82% missing NPI and 100% missing address — limitations of the PDF table structure, not conversion errors.

Massachusetts: 100% missing address — source Excel has no address column.

NPI coverage varies by state: MA ~81%, MS ~75%, MI ~29%, MT ~25%, NE ~20%, MD ~17%.

Claims matching for NE and MA will rely primarily on provider name where NPI or address is unavailable.

These limitations are documented in docs/ISSUES_AND_DECISIONS_BILINGUAL.md and docs/DATA_INVENTORY.md.

8. Data Quality and Regression Safety

Improvements

Added 55 pytest cases covering core transforms, dedup logic, and record-count regression against baselines.

Integrated `quality_audit` into the main pipeline with fail-fast on rerun inconsistency.

Tightened validation thresholds from loose minimums to expected counts ± 5 per state.

Added `verify_main_sync` Python assertion — pipeline exits with error if `exclusion_main` differs from `cleaned_staging`.

Run manifest tracks source file SHA256 on each pipeline run for reproducibility and audit.

9. GitHub Documentation and Reproducibility

Repository: <https://github.com/Xinzhuo-Li/medicaid-exclusion-list>

`README.md` — clone-to-run quick start (Path A full ETL, Path B CSV-only), pipeline overview, troubleshooting.

`docs/WORKFLOW.md` — detailed operator runbook with per-step manual commands and JSON report interpretation.

`data/raw/README.md` — exact source filenames and update instructions.

All seven source files (6 `xlsx` + 1 `pdf`) included in the repo so new users can reproduce the full pipeline after clone.

Pre-generated processed/cleaned CSVs included for validation and testing without re-converting.

10. Lessons Learned

State exclusion lists are not uniform — each state requires its own converter with documented mapping rules before any merge is possible.

Real ETL work goes beyond loading data: validation, dedup policy, business-rule decisions, and audit trails are essential for healthcare compliance use cases.

Leadership decisions on edge cases (empty dates, long names, cross-state NPIs) must be documented explicitly to avoid silent data loss.

Automated tests and rerun consistency checks catch regressions early when shared cleaning logic is changed.

Staging tables plus a strict merge pattern (DELETE scoped by state, INSERT from staging, EXCEPT verify) keeps exclusion_main trustworthy.

Clear GitHub documentation reduces onboarding time and helps the team continue development without depending on one person's local setup.

Database design and search requirements (NPI vs name search, source labeling, full business names) must be considered during ETL, not only in the UI.

11. Main Takeaways

Successfully delivered a production-ready six-state Medicaid exclusion list ETL with 8,575 records in PostgreSQL and full documentation on GitHub.

The project demonstrates the full data engineering workflow: ingest heterogeneous sources, normalize to a standard schema, validate quality, load to database, merge with audit, and document for the team.

Leadership-confirmed business rules are encoded in code, SQL, and bilingual decision documents — the system is ready for integration with the Django search application.

The foundation is in place to add more states, connect to the search UI, and continue data quality monitoring as source files are updated.

Deliverables

- GitHub repository: <https://github.com/Xinzhuo-Li/medicaid-exclusion-list>
- PostgreSQL: 8,575 rows in exclusion_main on vesta (port 5433)
- Documentation: README.md, WORKFLOW.md, DATA_INVENTORY.md, STATE_MAPPING.md, ISSUES_AND_DECISIONS_BILINGUAL.md
- Automated tests: 55 pytest cases in tests/
- Audit artifacts: validation reports, quality audit reports, dedup logs, run manifests in docs/

Future Improvements

- Add more states to the ETL pipeline using the established per-state converter pattern.

- Add GitHub Actions CI to run pytest and pipeline `--skip-db` on each push.
- Integrate `exclusion_main` with the Django search application (name, NPI, business name, state, source, exclusion date filters).
- Monitor source file updates via run manifest SHA256 diffs and refresh baselines when counts change intentionally.
- Consider Git LFS if future source files exceed GitHub size limits.
- Add Nebraska PDF regression fixture tests if PDF layout changes become a recurring risk.
- Extend deploy scripts for non-Mac environments (remove `osascript/expect` dependency where possible).
- Continue separating Medicare (federal/OIG) and Medicaid (state) source labels in the unified database as additional sources are added.

Hours Worked: 80